

## KUHMO KALLIOJOKI, Finland

WMO: 027990

Lat: 64.298N

Lon: 30.166E

Elev: 651

StdP: 14.35

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -24.5      | -18.7 | -30.7                         | 1.0 | -24.6 | -24.1 | 1.5 | -18.6 | 14.2                  | 27.6 | 13.0 | 26.5 | 1.0                   | 230  | 0.633 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 80.0            | 63.7 | 76.1 | 61.8 | 72.3 | 60.3 | 66.8                | 74.6 | 64.5 | 71.9 | 62.2 | 69.6 | 5.7                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.9                            | 91.2 | 70.4 | 61.6 | 83.9 | 67.4 | 59.3 | 77.2 | 65.1 | 31.7          | 74.6 | 29.9 | 71.7 | 28.2 | 70.2 | 73.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.0          | 11.6 | 10.1 | DB                         | -30.2 | 82.7               | 8.1 | 5.9                                                | -36.0 | 87.0       | -40.8 | 90.4       | -45.3 | 93.7       | -51.2 | 98.0 | (4) |
| (5)               |      |      | WB                         | -30.0 | 68.5               | 7.8 | 3.5                                                | -35.6 | 71.0       | -40.1 | 73.1       | -44.5 | 75.1       | -50.1 | 77.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 35.5        | 10.3  | 14.2  | 21.0  | 32.9  | 45.9  | 53.7  | 60.5  | 56.5  | 47.9  | 35.6  | 27.2  | 18.9  |  |  |
|                                                                        | DBStd       | 19.22       | 14.70 | 13.94 | 10.88 | 8.01  | 8.78  | 7.29  | 6.80  | 6.22  | 5.99  | 6.66  | 8.69  | 12.84 |  |  |
|                                                                        | HDD50       | 6099        | 1232  | 1003  | 898   | 513   | 185   | 42    | 5     | 16    | 110   | 447   | 684   | 964   |  |  |
|                                                                        | HDD65       | 10819       | 1697  | 1423  | 1363  | 962   | 593   | 345   | 175   | 273   | 514   | 911   | 1134  | 1429  |  |  |
|                                                                        | CDD50       | 805         | 0     | 0     | 0     | 1     | 57    | 154   | 330   | 216   | 46    | 1     | 0     | 0     |  |  |
|                                                                        | CDD65       | 50          | 0     | 0     | 0     | 0     | 1     | 7     | 34    | 8     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 549         | 0     | 0     | 0     | 0     | 43    | 109   | 314   | 83    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 87          | 0     | 0     | 0     | 0     | 0     | 14    | 59    | 14    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.8         | 4.2   | 4.9   | 5.2   | 4.8   | 5.3   | 4.8   | 4.1   | 4.2   | 4.6   | 5.3   | 5.2   | 4.9   |  |  |
|                                                                        | PrecAvg     | 24.0        | 1.5   | 1.2   | 1.2   | 1.3   | 2.0   | 2.4   | 2.9   | 3.1   | 2.4   | 2.0   | 1.9   | 1.9   |  |  |
| (15) Precipitation                                                     | PrecMax     | 32.1        | 3.0   | 2.3   | 2.2   | 2.7   | 3.4   | 3.9   | 5.7   | 5.4   | 4.0   | 4.8   | 3.6   | 4.3   |  |  |
|                                                                        | PrecMin     | 17.3        | 0.4   | 0.4   | 0.5   | 0.3   | 0.6   | 1.0   | 1.5   | 1.2   | 0.6   | 0.2   | 0.7   | 0.3   |  |  |
|                                                                        | PrecStd     | 4.0         | 0.7   | 0.6   | 0.4   | 0.6   | 0.9   | 0.8   | 1.2   | 1.2   | 1.0   | 1.2   | 0.7   | 1.0   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 35.4  | 38.5  | 45.2  | 61.9  | 78.6  | 82.2  | 84.7  | 81.7  | 68.1  | 53.1  | 45.1  | 36.8  |  |  |
|                                                                        |             | MCWB        | 35.1  | 35.8  | 37.6  | 47.8  | 56.7  | 62.9  | 68.7  | 65.7  | 59.4  | 50.3  | 43.8  | 36.3  |  |  |
|                                                                        | 2%          | DB          | 32.9  | 35.1  | 39.7  | 53.8  | 74.1  | 77.4  | 81.3  | 75.5  | 62.8  | 49.6  | 41.9  | 34.7  |  |  |
|                                                                        |             | MCWB        | 32.5  | 33.9  | 34.2  | 43.8  | 55.8  | 61.9  | 65.4  | 62.8  | 55.1  | 47.3  | 41.4  | 34.2  |  |  |
|                                                                        | 5%          | DB          | 31.3  | 33.1  | 36.5  | 49.0  | 68.2  | 71.9  | 77.4  | 71.4  | 59.9  | 46.9  | 38.8  | 33.4  |  |  |
|                                                                        |             | MCWB        | 31.0  | 32.4  | 33.0  | 40.0  | 53.7  | 58.5  | 64.0  | 60.6  | 54.5  | 45.2  | 38.3  | 33.0  |  |  |
|                                                                        | 10%         | DB          | 29.3  | 31.7  | 34.4  | 45.2  | 62.3  | 67.8  | 73.6  | 67.7  | 57.2  | 44.6  | 36.5  | 32.3  |  |  |
|                                                                        |             | MCWB        | 29.0  | 31.3  | 32.1  | 37.8  | 50.2  | 56.1  | 62.2  | 59.6  | 53.3  | 43.3  | 36.1  | 32.0  |  |  |
|                                                                        | 0.4%        | WB          | 34.9  | 36.4  | 38.2  | 49.4  | 59.8  | 67.1  | 70.9  | 68.0  | 60.5  | 51.3  | 44.7  | 36.7  |  |  |
|                                                                        |             | MCDB        | 35.3  | 37.4  | 43.0  | 57.6  | 71.7  | 74.3  | 82.2  | 78.2  | 66.1  | 52.5  | 44.9  | 36.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 32.4  | 34.2  | 35.5  | 44.7  | 57.3  | 63.8  | 68.1  | 64.2  | 57.5  | 47.7  | 41.3  | 34.2  |  |  |
|                                                                        |             | MCDB        | 32.6  | 35.0  | 38.2  | 53.5  | 68.3  | 73.8  | 75.8  | 70.9  | 61.1  | 48.6  | 41.8  | 34.4  |  |  |
|                                                                        | 5%          | WB          | 30.8  | 32.6  | 34.0  | 40.7  | 54.6  | 60.8  | 66.0  | 62.3  | 55.4  | 45.4  | 38.5  | 33.1  |  |  |
|                                                                        |             | MCDB        | 31.0  | 33.1  | 35.7  | 46.5  | 66.9  | 68.9  | 73.6  | 69.2  | 58.3  | 46.5  | 38.9  | 33.3  |  |  |
|                                                                        | 10%         | WB          | 28.6  | 31.1  | 32.7  | 38.8  | 51.3  | 58.1  | 63.8  | 60.6  | 53.8  | 43.4  | 36.1  | 32.0  |  |  |
|                                                                        |             | MCDB        | 28.9  | 31.7  | 34.2  | 43.0  | 60.9  | 65.0  | 71.0  | 66.5  | 56.5  | 44.5  | 36.3  | 32.3  |  |  |
|                                                                        | 5% DB       | MDBR        | 12.6  | 14.1  | 18.0  | 17.6  | 18.9  | 17.8  | 17.2  | 16.1  | 12.7  | 8.4   | 7.2   | 9.5   |  |  |
|                                                                        |             | MCDBR       | 7.1   | 8.7   | 15.0  | 24.1  | 28.9  | 25.3  | 22.9  | 23.0  | 17.2  | 11.4  | 7.3   | 6.6   |  |  |
|                                                                        |             | MCWBR       | 7.3   | 8.2   | 11.5  | 14.2  | 13.2  | 11.5  | 9.5   | 11.0  | 10.5  | 9.6   | 7.4   | 6.3   |  |  |
|                                                                        |             | MCWBR       | 7.2   | 9.0   | 11.5  | 20.6  | 25.9  | 21.0  | 20.5  | 18.8  | 14.3  | 10.5  | 7.4   | 6.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR       | 7.3   | 8.6   | 9.5   | 12.7  | 12.9  | 10.8  | 9.7   | 9.9   | 9.4   | 9.4   | 7.3   | 6.2   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        | taub        |             | 0.209 | 0.274 | 0.303 | 0.337 | 0.345 | 0.342 | 0.367 | 0.346 | 0.323 | 0.300 | 0.359 | 0.202 |  |  |
|                                                                        |             | taud        | 1.949 | 2.101 | 2.152 | 2.257 | 2.453 | 2.487 | 2.435 | 2.475 | 2.526 | 2.386 | 2.657 | 1.953 |  |  |
| (40) Clear-Sky Solar Irradiance                                        | Ebn at Noon |             | 108   | 197   | 245   | 264   | 275   | 278   | 266   | 262   | 245   | 194   | 102   | 40    |  |  |
|                                                                        |             | Edh at Noon | 9     | 20    | 29    | 33    | 30    | 30    | 31    | 27    | 21    | 15    | 8     | 3     |  |  |
|                                                                        | RadAvg      |             | 47    | 221   | 688   | 1193  | 1466  | 1631  | 1531  | 1142  | 637   | 227   | 52    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadStd      |             | 5     | 45    | 69    | 113   | 173   | 109   | 166   | 110   | 70    | 34    | 10    | 2     |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (22 neighbors) |  | +0.97 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page